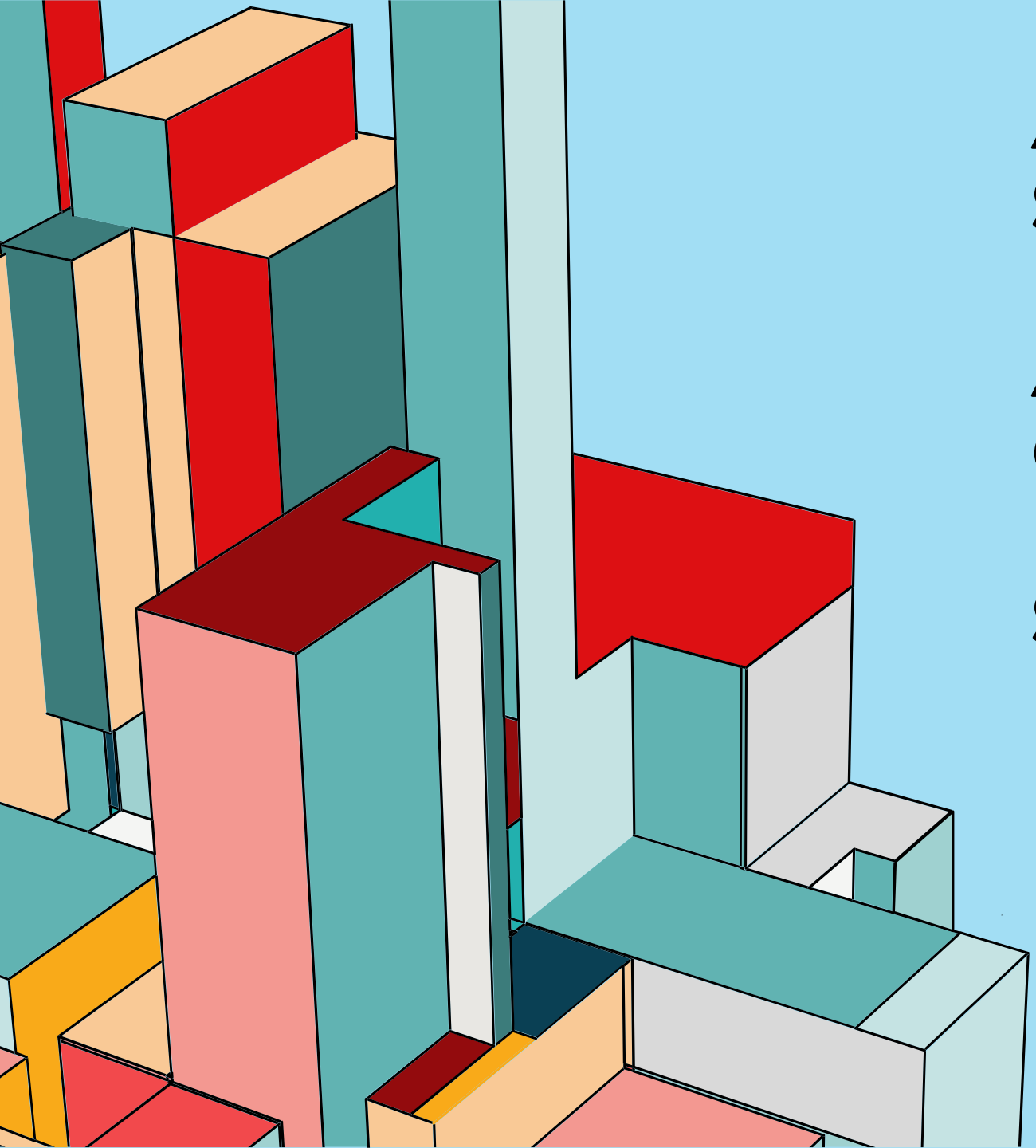


APPLIED DATA SCIENCE PROJECT

ARTWORK TYPE CLASSIFICATION FOR METADATA SUGGESTION

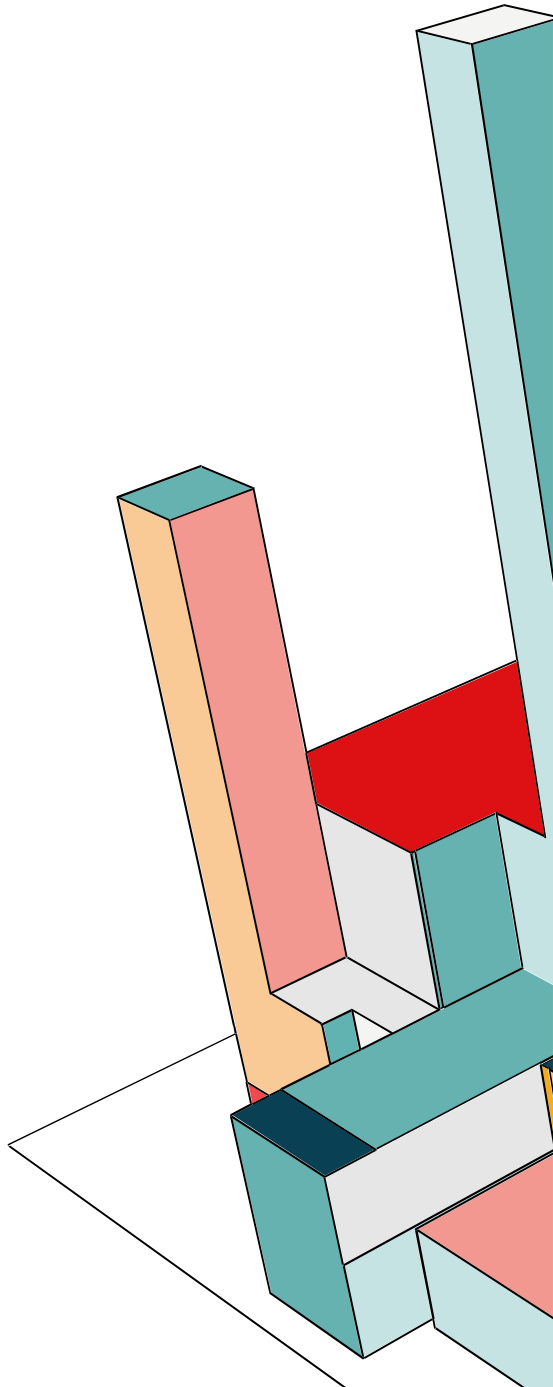
Politecnico di Torino

FONDAZIONE links PASSION FOR INNOVATION

ellis European Laboratory for Learning and Intelligent Systems

VALUE-DRIVEN PROJECT

- The **digitisation of cultural heritage** produces large collections of artefacts that curators must manually annotate with metadata, a slow and costly process
- Within the **REEVALUATE** project, a **metadata suggestion pipeline** supports curators by proposing metadata for newly digitised artefacts starting from their images
- This project aims to develop an **efficient** machine learning model that automatically **classifies the type of artefact** from its image, to be integrated in the pipeline to improve its suggestions



DATA

- ARTKB Knowledge Base:

- Digitised artefacts with curated metadata

~150,000

images of digitised artefacts

Example artefact types

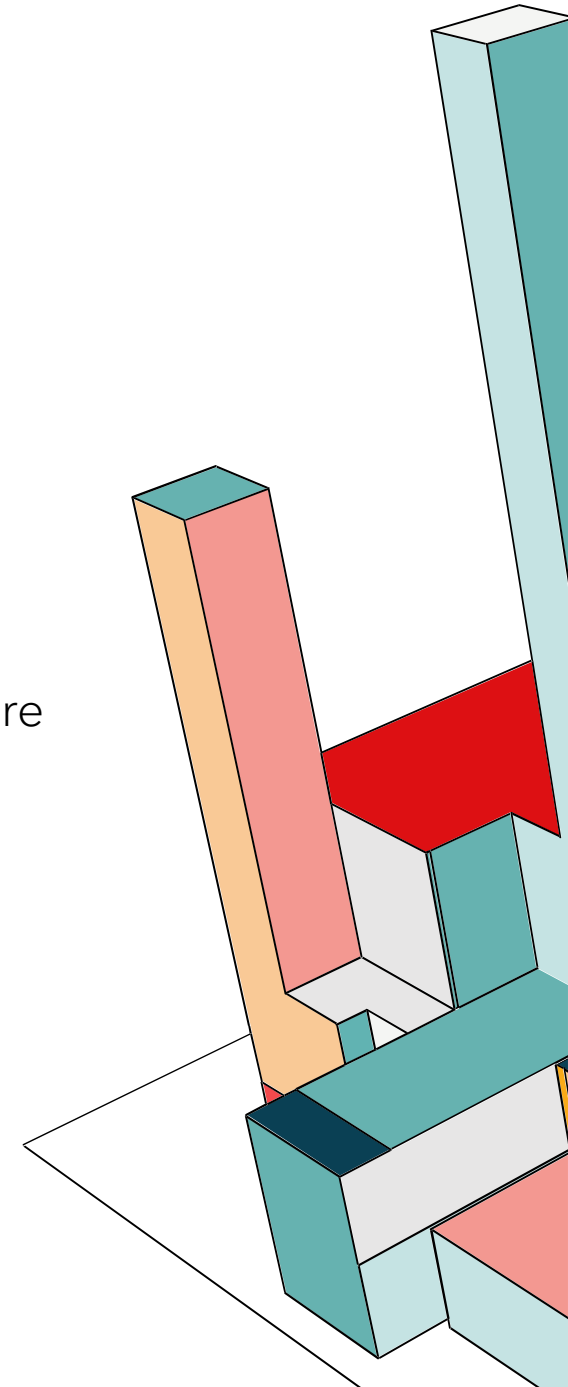
Painting | Sculpture | Clothing | ...

- With annotations of

Artefact type

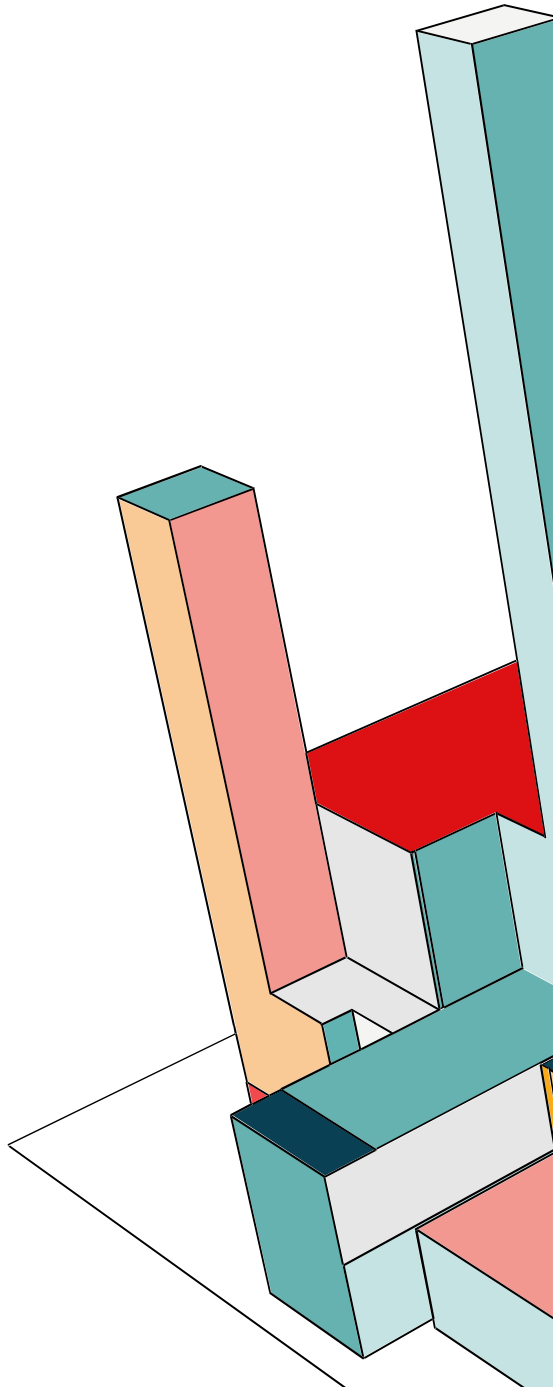
Material, movement, genre

Subject tags

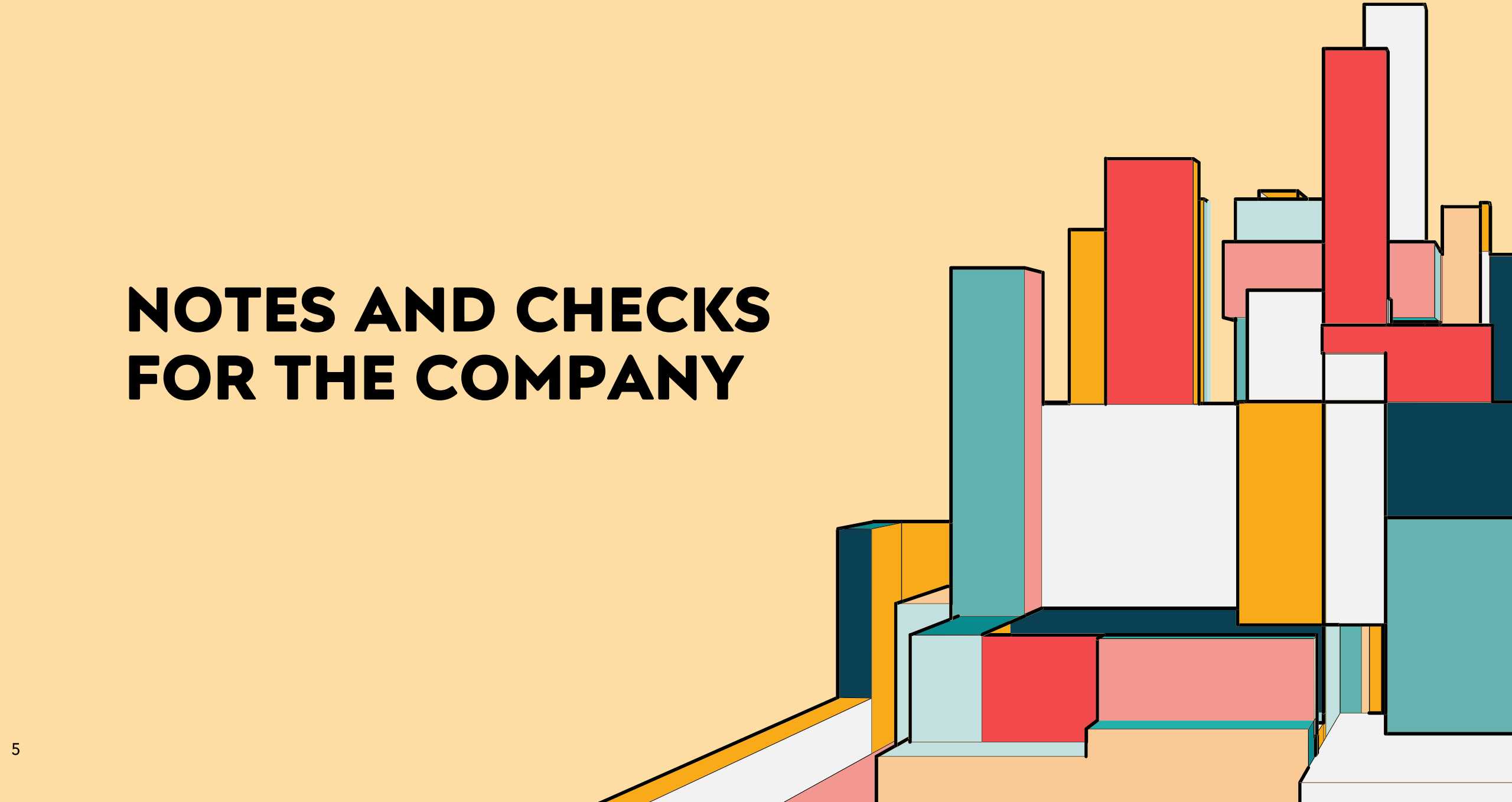


TASK

- **Image Classification** task where the objective is to predict the **type of artefact** depicted in an image, according to a defined taxonomy
- Participants are tasked to develop an **efficient machine-learning** model that processes a single **image** and outputs its artefact type, suitable for integration in the metadata suggestion pipeline
- This is a **new task**: no previous baseline is available, so participants will establish the reference results
- Reference metrics are **Accuracy**, **Top-k Accuracy** and **macro F1-score**, together with **efficiency** measures (model size, inference time)

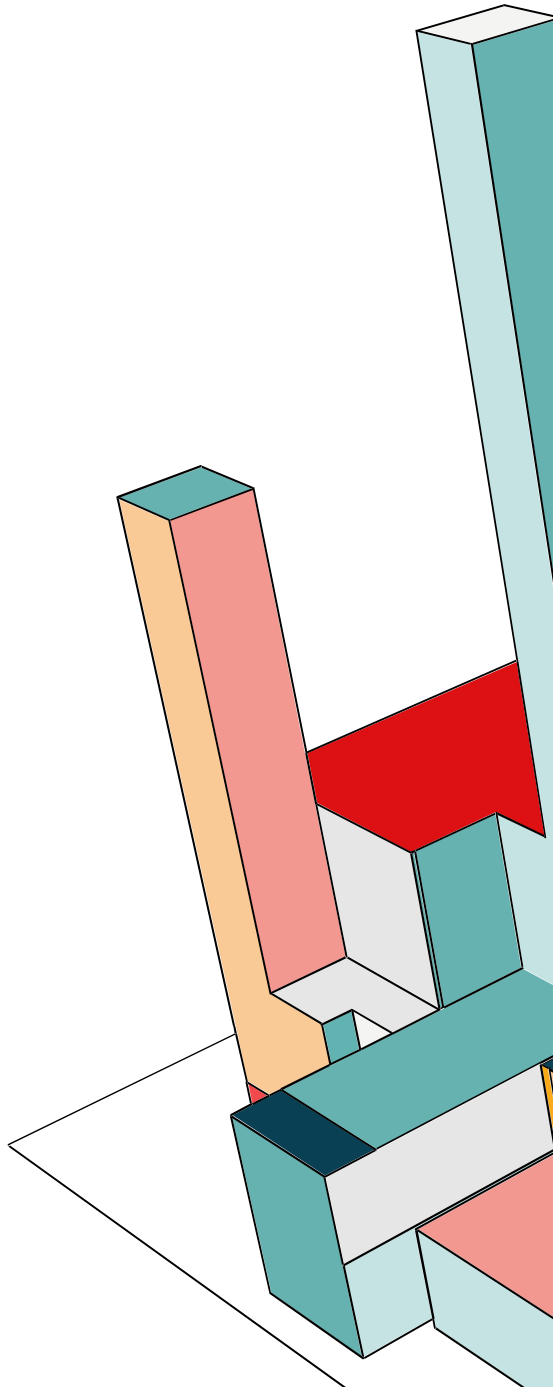


NOTES AND CHECKS FOR THE COMPANY



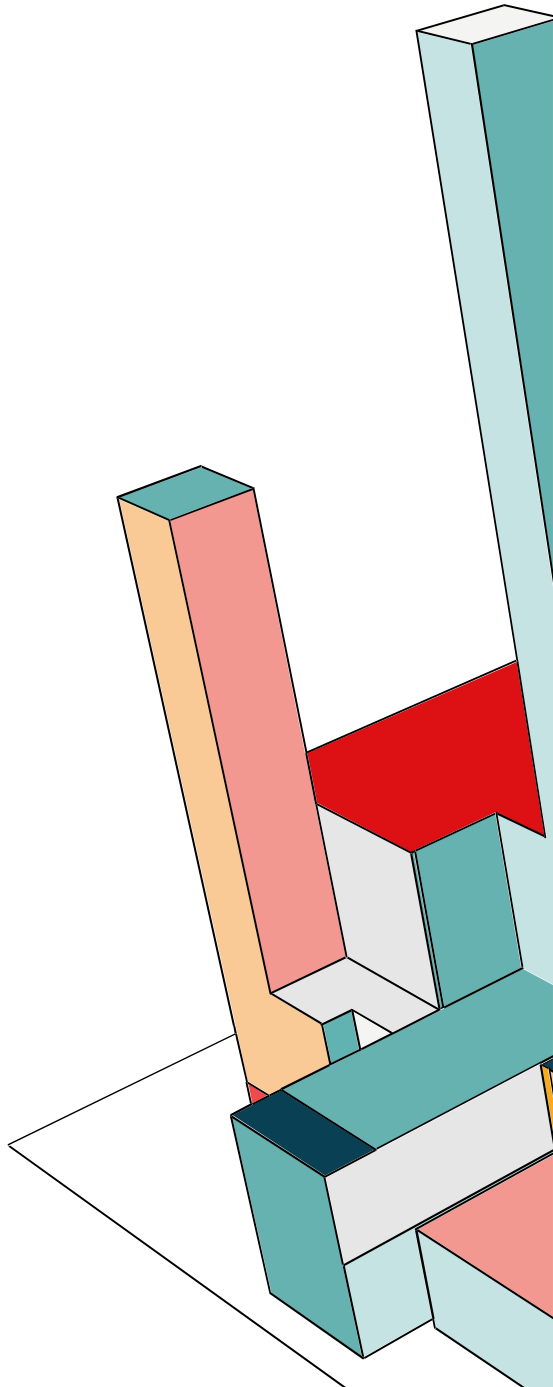
LIGHT MENTORING

- The following meetings will be organized:
    - **First (longer) meeting** as introduction to the existing metadata suggestion pipeline and methodology, context and data presentation.
    - **Periodic bi-weekly meetings** (1h) with LINKS to align with the current development.



POLICY

- Data are open and publicly available
- No NDA agreement is required for the provided data
- Code and methodology could be published



THANKS FOR THE ATTENTION

CONTACTS

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